Supplementary Video 1

Toroidal envelope, stability boundary, vortex structure and worldline geometry

Manuscript: Displacement-Engineered Warp Fields: Stability Surfaces and Worldline Geometry.

The silent animation shows an original schematic redraw of nested toroidal surfaces labeled E(D), a boundary labeled $C(D, \nabla D, \nabla^2 D, I_a) = 0$, and the field-level labels $|\psi|^2 = 0.5$ and $|\xi|^2 = 0.03$. A rotating torus carries qualitative vortex-magnitude shading and moving markers. A family of straight curves is interpolated into curved paths, followed by a diagram of the identity-space rate relation $dI_a/d\tau = f(D)$.

The surfaces and curves are figure-based reconstructions. The video contains no sampled displacement field, evaluated stability functional, numerical curl values, integrated worldline solution or numerical identity-space trajectory. Color shading is qualitative and motion is display interpolation.

Duration: 36 seconds. Frame rate: 24 frames per second. Resolution: 1280 × 720 pixels. Format: MP4; H.264; YUV 4:2:0. Audio: none.

Accessibility description

A silent animation rotates nested toroidal surfaces, traces a labeled boundary, shows qualitative vortex shading, bends a family of nonintersecting curves, and finishes with a diagram linking geometric flow to an identity-space rate. On-screen labels identify each relation; a separate timed visual transcript accompanies the video.

File: JMP26-AR-02831-SuppVideo1.mp4